

✓ AI Product Description Generator

Project 4 — AI Automation Portfolio

This project uses Python, Prompt Engineering, and the OpenAI API to generate clear, engaging, and conversion-focused product descriptions.

The AI creates product marketing content using:

- Product name
- Product category
- Product features
- Target audience
- Brand tone
- Main selling goal

Project Goal

The goal of this project is to automate the product-copywriting process by converting structured product information into clear, accurate, audience-focused, and e-commerce-ready product descriptions.

✓ Step 2.1: Create Product Inputs and the AI Prompt

Prompt Design and Core Logic

In this section, we define the product and marketing information required to generate an accurate, engaging, and e-commerce-ready product description.

```
# Product and marketing information

product_name = "EcoSip Reusable Water Bottle"

product_category = "Sustainable drinkware"

product_features = """
- Made from stainless steel
- Double-wall insulated design
- Leak-resistant lid
- 750 ml capacity
- Reusable and designed for everyday use
```

```
"""

target_audience = """
Environmentally conscious students and working professionals
who want a practical reusable water bottle
"""

brand_tone = "Friendly, modern, and trustworthy"

selling_goal = """
Encourage customers to purchase the product from the online store
"""
```

```
# Create a structured prompt for AI product-description generation

prompt = f"""
You are an expert e-commerce copywriter, product marketer,
and SEO content specialist.

Your task is to create an accurate, engaging, audience-focused,
and conversion-oriented product description using only the
product information provided below.

Product Name:
{product_name}

Product Category:
{product_category}

Product Features:
{product_features}

Target Audience:
{target_audience}

Brand Tone:
{brand_tone}

Primary Selling Goal:
{selling_goal}

Follow these instructions carefully:

1. Use only the product information provided.
2. Do not invent specifications, materials, certifications,
   warranties, discounts, customer reviews, or product results.
3. Explain how the listed features may be useful to the target audience.
4. Use the requested brand tone consistently.
5. Write clear, natural, and customer-friendly language.
6. Avoid exaggerated or unsupported claims.
7. Focus on relevant benefits without changing the product facts.
8. Make the content suitable for an e-commerce product page.
9. Include a clear but natural call to action.
```

```
10. Suggest relevant SEO keywords based only on the provided information.
```

```
Use exactly this output structure:
```

```
## Product Title
```

```
## Short Description
```

```
## Detailed Product Description
```

```
## Key Features
```

```
## Customer Benefits
```

```
## Call to Action
```

```
## Suggested SEO Keywords
```

```
"""
```

```
print(prompt)
```

```
EcoSip Reusable Water Bottle
```

```
Product Category:  
Sustainable drinkware
```

```
Product Features:
```

- Made from stainless steel
- Double-wall insulated design
- Leak-resistant lid
- 750 ml capacity
- Reusable and designed for everyday use

```
Target Audience:
```

```
Environmentally conscious students and working professionals  
who want a practical reusable water bottle
```

```
Brand Tone:  
Friendly, modern, and trustworthy
```

```
Primary Selling Goal:
```

```
Encourage customers to purchase the product from the online store
```

```

5. Write clear, natural, and customer-friendly language.
6. Avoid exaggerated or unsupported claims.
7. Focus on relevant benefits without changing the product facts.
8. Make the content suitable for an e-commerce product page.
9. Include a clear but natural call to action.
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```

Use exactly this output structure:

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## Product Title
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## Short Description
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```
## Detailed Product Description
```

```
## Key Features
```

```
## Customer Benefits
```

```
## Call to Action
```

```
## Suggested SEO Keywords
```

✓ Step 2.2: AI-Powered Product Description Generation

In this section, the structured product prompt is sent to the OpenAI API. The AI analyzes the product information and generates an accurate, engaging, audience-focused, and e-commerce-ready product description.

```
!pip install -q openai
```

```

from google.colab import userdata

api_key = userdata.get("OPENAI_API_KEY")

```

```

from openai import OpenAI

client = OpenAI(api_key=api_key)

```

```

# gerate the product description
response = client.responses.create(
    model="gpt-5-mini",
    input=prompt
)

product_description = response.output_text

print(product_description)

```

Quality Check

Review the generated output and confirm:

- The product title is relevant to EcoSip.
- The short description is clear and engaging.
- The detailed description uses the provided features.
- Stainless steel is mentioned accurately.
- The double-wall insulated design is included.
- The leak-resistant lid is included.
- The 750 ml capacity is included.
- The description is suitable for students and working professionals.
- The tone is friendly, modern, and trustworthy.
- The call to action is natural.
- The SEO keywords are relevant.
- No unsupported specifications or claims were invented.

Pay special attention to unsupported claims.

The AI should not add details such as:

- "Keeps drinks cold for 24 hours"
- "BPA-free"
- "Dishwasher-safe"
- "Lifetime warranty"
- "Certified sustainable"

unless those details were included in your product information.

✓ Test a New Product Scenario

We'll test the generator using a wireless desk lamp.

Part A

— Replace the Product Inputs

```
# Product and marketing information

product_name = "LumaFlex Wireless Desk Lamp"

product_category = "Rechargeable desk lighting"

product_features = ""
- Wireless and rechargeable design
- Adjustable brightness levels
```

```
- Flexible lamp neck
- Compact size
- USB charging cable included
"""

target_audience = """
Students, remote workers, and professionals who need
a practical and flexible lighting solution for desks
and workspaces
"""

brand_tone = "Modern, professional, and helpful"

selling_goal = """
Increase product awareness and encourage customers
to purchase from the online store
"""
```

✓ Generate the New Product Description

```
response = client.responses.create(
    model="gpt-5-mini",
    input=prompt
)

product_description = response.output_text

print(product_description)
```

✓ Refine the Prompt

```
# Refined prompt for AI product-description generation

prompt = f"""
You are an expert e-commerce copywriter, product marketer,
SEO content specialist, and customer-focused product writer.

Your task is to create an accurate, clear, engaging, and
conversion-oriented product description using only the
product information provided below.

PRODUCT INFORMATION

Product Name:
{product_name}

Product Category:
{product_category}
```

Product Features:
{product_features}

TARGET CUSTOMER

Target Audience:
{target_audience}

BRAND REQUIREMENTS

Brand Tone:
{brand_tone}

Primary Selling Goal:
{selling_goal}

Follow these rules carefully:

1. Use only the product facts provided.
2. Do not invent specifications, dimensions, materials, certifications, warranties, discounts, prices, customer reviews, ratings, or product results.
3. Do not claim that the product is the "best," "perfect," "guaranteed," or superior to competitors.
4. Clearly distinguish between product features and customer benefits.
5. Explain benefits only when they are reasonably supported by the listed features.
6. Write for the specified target audience.
7. Use the requested brand tone consistently.
8. Keep the language clear, natural, and easy to scan.
9. Avoid unnecessary repetition and generic marketing phrases.
10. Make the content suitable for an e-commerce product page.
11. Use concise paragraphs and readable bullet points.
12. Include a clear, natural, and non-misleading call to action.
13. Suggest SEO keywords based only on the provided product name, category, features, and intended use.
14. Do not add facts that were not provided.

Use exactly this output structure:

Product Title

Create a clear and descriptive product title.
Do not add unsupported features.

Short Description

Write 2 to 3 concise sentences.

Detailed Product Description

```
Write 1 to 2 clear paragraphs that explain the product,
its main features, and its relevant uses.
```

```
## Key Features
```

```
List the provided product features as clear bullet points.
```

```
## Customer Benefits
```

```
List practical benefits supported by the provided features.
```

```
## Call to Action
```

```
Write one concise and natural call to action.
```

```
## Suggested SEO Keywords
```

```
Provide 8 to 12 relevant keywords or keyword phrases.
"""
```

✓ Interactive Product Description Generator

This section allows users to enter product and marketing information and generate a customized AI-powered product description.

```
# Collect product and marketing information from the user

product_name = input("Enter the product name: ")

product_category = input("Enter the product category: ")

product_features = input(
    "Enter the main product features (separate features with commas): "
)

target_audience = input("Describe the target audience: ")

brand_tone = input(
    "Enter the desired brand tone (for example: friendly, modern, professional): "
)

selling_goal = input("Enter the primary selling goal: ")

print("\nProduct information received successfully.")
```

✓ Generate the Interactive Product Description


```
response = client.responses.create(  
    model="gpt-5-mini",  
    input=prompt  
)  
  
product_description = response.output_text  
  
print(product_description)
```

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Quality Check

Review the generated output and confirm:

- The product is identified correctly.
- The description is relevant to portable coffee preparation.
- The compact design is included.
- Manual operation is included.
- The reusable filter is included.
- The lightweight design is included.
- Travel and outdoor use are represented.
- The target audience is considered.
- The tone is warm, practical, and adventurous.
- The description is easy to scan.
- Features and benefits are clearly separated.
- The call to action is natural.
- The SEO keywords are relevant.
- No unsupported product claims are included.

Be especially careful that the AI does not invent details such as:

- Brewing time
- Coffee capacity
- Battery operation
- Dishwasher safety
- Included accessories
- Temperature control
- Product warranty

unless those facts were provided.

Quality Check

Review the generated output and confirm:

- The product is identified correctly.
- The description is relevant to portable coffee preparation.
- The compact design is included.
- Manual operation is included.
- The reusable filter is included.
- The lightweight design is included.
- Travel and outdoor use are represented.
- The target audience is considered.
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Be especially careful that the AI does not invent details such as:

- Brewing time
- Coffee capacity
- Battery operation
- Dishwasher safety
- Included accessories

- Temperature control
- Product warranty

unless those facts were provided.